2Fast2Detections:

Joint Detection of AI-Generated Images and Post-Processing Alterations in Real-World Scenarios

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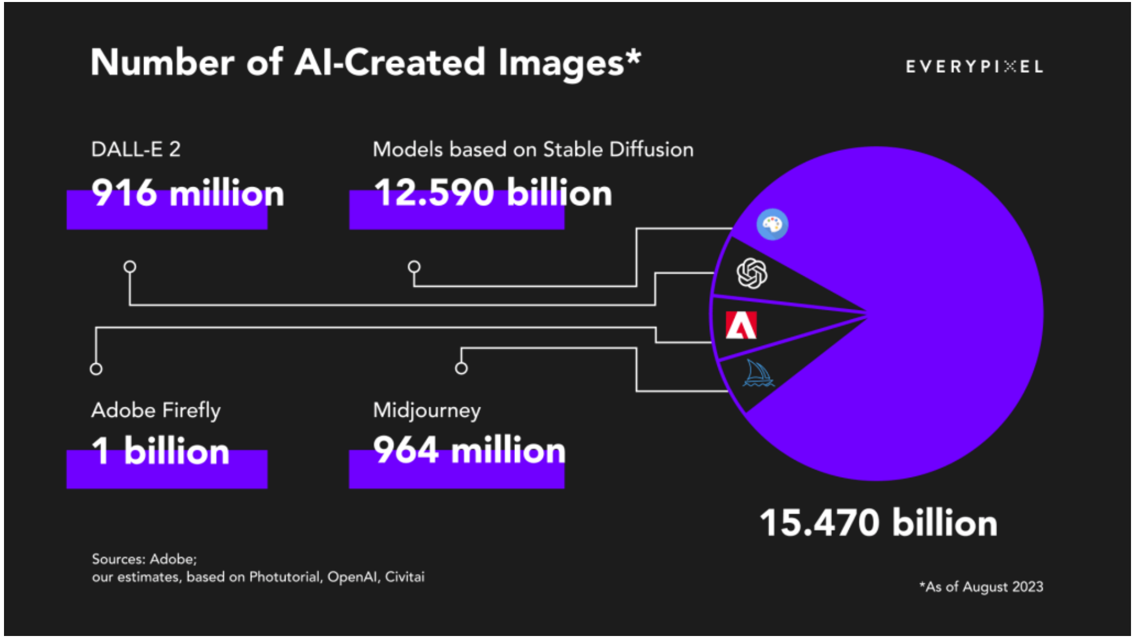
Problem Description: Growth of AI Images



Source: Everypixel (2024)

Everypixel Journal. (2023, August). *15 billion images created by AI in just one year*. Everypixel. https://journal.everypixel.com/ai-image-statistics

Problem Description & State of the Art

Disinformation

Scientific Fraud

Identity Theft & Deepfakes

Risks of AI generated Images

Disinformation

AI-generated images can trigger rapid misinformation events, creating market panic and social confusion when fakes are indistinguishable from real visuals. In 2023 **30% of content-manipulated misinformation images**.

Source: Dufour et al. (2024)

Source: Reuters (2026)

Dufour, N., Pathak, A., Samangouei, P., Hariri, N., Deshetti, S., Dudfield, A., Guess, C., Escayola, P. H., Tran, B., Babakar, M., & Bregler, C. (2024). AMMeBa: A large-scale survey and dataset of media-based misinformation in-the-wild. *arXiv*. https://doi.org/10.48550/arxiv.2405.11697

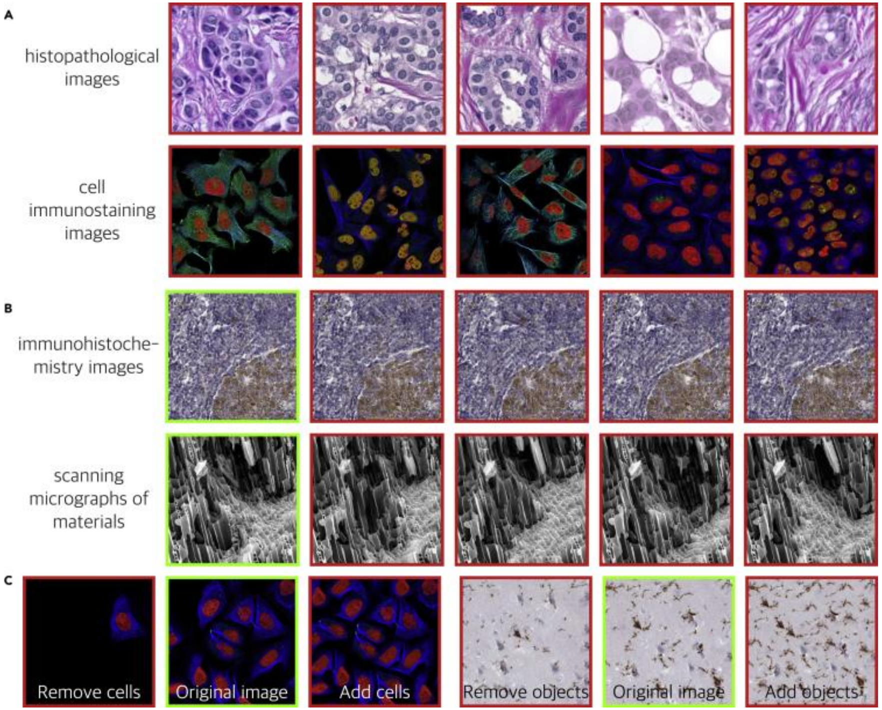
Reuters Fact Check. (2026, January 6). *Fact check: Photo of Maduro in black jacket among uniformed personnel is AI-generated*. Reuters. https://www.reuters.com/fact-check/photo-maduro-black-jacket-among-uniformed-personnel-is-ai-generated-2026-01-06/

Risks of AI generated Images

Scientific Fraud

Fabrication of research data and imagery, threatening the integrity of academic publishing and scientific trials.

Source: Gu et al. (2024)



Source: Gu et al. (2024)

Gu, J., Wang, X., Li, C., Zhao, J., Fu, W., Liang, G., & Qiu, J. (2022). AI-enabled image fraud in scientific publications. *Patterns*, *3*, 100511. https://doi.org/10.1016/j.patter.2022.100511

Risks of AI generated Images

Identity Theft & Deepfakes

non-consensual fake images/videos of real people used for harassment, fraud, or reputational harm.

Source: NBC. (2023)

Tolentino, D. (2023, March 27). *AI-generated images of Pope Francis in puffer jacket fool the internet*. NBC News. https://www.nbcnews.com/tech/pope-francis-ai-generated-images-fool-internet-rcna76838

Techniques for Identifying AI generated Images

1 CNN artifact detection

Examples: ResNet-50, CNNDetection, Gram-Net, LGrad

- Classic image classifiers trained to spot low-level artifacts, textures, gradients, and local pixel inconsistencies
- Strong on clean or familiar data, but overfit to the training distribution

Learns pixel-level clues, but can fail when images come from new generators or are compressed, resized, or reposted - accuracy often below 55% on unseen models.

Tan et al. (2023), Liu et al. (2020), Wang et al.(2020), He et al.(2016)

2 Frequency-based detection

Examples: FreqNet, DFFreq

- Analyzes the image's frequency spectrum for unnatural patterns rather than relying only on visible pixels
- Shows stronger generalization across generators than CNN artifact methods

Searches for hidden spectral artifacts left by generative models, often invisible to humans - a strong trade-off between real- and fake-image accuracy under robustness tests.

Yan et al. (2025), Tan et al. (2025)

3 CLIP-based detection

Examples: CLIPDetection, AIDE

- Uses large pre-trained vision-language models (e.g. CLIP-ViT) to represent images in a rich, general feature space
- Fixed embeddings with a lightweight classifier on top distinguish real vs. fake

Uses large pre-trained vision models to spot high-level visual patterns that may generalize beyond a single generator - best generalization of the three.

Yan et al. (2025), Ojha et al. (2023)

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778.
Liu, Z., Qi, X., & Torr, P. H. S. (2020). Global texture enhancement for fake face detection in the wild. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 8060–8069.
Ojha, U., Li, Y., & Lee, Y. J. (2023). Towards universal fake image detectors that generalize across generative models. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 24480–24489.
Tan, C., Zhao, Y., Wei, S., Gu, G., & Wei, Y. (2023). Learning on gradients: Generalized artifacts representation for GAN-generated images detection. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 12105–12114.
Tan, C., Zhao, Y., Wei, S., Gu, G., Liu, P., & Wei, Y. (2024). Frequency-aware deepfake detection: Improving generalizability through frequency space domain learning. Proceedings of the AAAI Conference on Artificial Intelligence, 38(5), 5052–5060.
Wang, S.-Y., Wang, O., Zhang, R., Owens, A., & Efros, A. A. (2020). CNN-generated images are surprisingly easy to spot... for now. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 8695–8704.
Yan, J., Li, Z., He, Z., & Fu, Z. (2025). Generalizable deepfake detection via effective local-global feature extraction. arXiv preprint arXiv:2501.15253.

Techniques for Identifying AI generated Images

Bridging the Gap Between Ideal and Real-world Evaluation: Benchmarking AI-Generated Image Detection in Challenging Scenarios

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Abstract

With the rapid advancement of generative models, highly realistic image synthesis has posed new challenges to digital security and media credibility. Although AI-generated image detection methods have partially addressed these concerns, a substantial research gap remains in evaluating their performance under complex real-world conditions. This paper introduces the Real-World Robustness Dataset (RRDataset) for comprehensive evaluation of detection models across three dimensions: 1) ***Scenario Generalization*** *– RRDataset encompasses high-quality images from seven major scenarios (War & Conflict, Disasters & Accidents, Political & Social Events, Medical & Public Health, Culture & Religion, Labor & Production, and everyday life), addressing existing dataset gaps from a content perspective. 2)* ***Internet Transmission Robustness*** *– examining detector performance on images that have undergone multiple rounds of sharing across various social media platforms. 3)* ***Re-digitization Robustness*** *– assessing model effectiveness on images altered through four distinct re-digitization methods.*

1. Introduction

With the emergence of powerful generative models [24, 40, 44, 45], AI-generated images are increasingly difficult to distinguish from real ones. Such indistinguishable AI-generated images pose risks to society, including the spread of misinformation and misleading visual cues. Consequently, the identification of AI-generated images has become a critical task with profound real-world implications.

An increasing number of studies explore various detection methods based on different features, including image texture [33], gradients [48], patch-level features [6, 9, 35], frequency-domain characteristics [12, 13, 15, 59], fine-grained image details [20, 46], and reconstruction loss [43, 54]. Vision-language models (VLMs) have also been adapted for detection tasks [25, 26, 38, 47]. Although conventional benchmarks such as GenImage [62], Fake2M [34], WildFake [18], and Chameleon [56] offer sizeable collections of generative images, they share two significant limitations. First, they primarily consist of everyday-life images, making it hard to assess whether the detection methods can effectively generalize to other scenarios. Sec-

Source: Li et al. (2025)

Model	Original		Transmission		Re-digitization		Overall
	Fake(%)	Real(%)	Fake(%)	Real(%)	Fake(%)	Real(%)	ACC(%)
Detectors (Train on GenImage-SDv1.4 & fine-tune on RRDataset)							
DRCT-ConvB[8]	93.52	95.52	92.82(-0.70)	95.09(-0.43)	64.34(-29.18)	96.22(+0.70)	89.59
DIRE[54]	89.72	98.25	90.34(+0.62)	97.87(-0.38)	1.42(-88.30)	98.89(+0.64)	79.42
DNF[60]	90.62	99.05	90.98(+0.36)	94.45(-4.60)	0.05(-90.57)	99.98(+0.93)	79.19
AIDE[56]	78.95	78.94	74.72(-4.23)	78.75(-0.19)	76.04(-2.91)	83.13(+4.19)	78.42
GramNet[33]	81.34	74.65	79.49(-1.85)	75.69(+1.04)	79.45(-1.89)	62.02(-12.63)	75.44
CNNSpot[53]	72.42	89.09	65.72(-6.70)	88.78(-0.31)	43.12(-29.30)	86.72(-2.37)	74.31
LNP[32]	83.14	89.26	38.23(-44.91)	89.30(+0.04)	31.91(-51.23)	91.05(+1.79)	70.48
Fredect[15]	79.60	75.93	58.13(-21.47)	82.11(+6.18)	46.34(-33.26)	69.95(-5.98)	68.68
NPR[51]	49.21	96.18	28.08(-21.13)	97.13(+0.95)	38.65(-10.56)	92.42(-3.76)	66.95
Fusing[20]	87.24	92.46	7.38(-79.86)	99.04(+6.58)	30.79(-56.45)	73.97(-18.49)	65.15
SAFE[29]	98.29	88.22	0.88(-97.41)	98.85(+10.63)	2.29(-96.00)	98.82(+10.60)	64.56
Freq-Net[50]	76.08	82.18	4.47(-71.61)	98.64(+16.46)	37.48(-38.60)	77.99(-4.19)	62.81
F3Net[55]	65.82	71.35	52.18(-13.64)	75.49(+4.14)	31.16(-34.66)	74.85(+3.50)	61.81
UnivFD[38]	64.79	64.90	44.61(-20.18)	70.80(+5.90)	36.15(-28.64)	75.69(+10.79)	59.49
C2P-CLIP[49]	17.21	97.54	28.82(+11.61)	99.58(+2.04)	18.01(+0.80)	90.29(-7.25)	58.58
SSP[9]	61.29	64.54	40.62(-20.67)	70.33(+5.79)	32.58(-28.71)	79.94(+15.40)	58.22
LGrad[48]	51.00	81.29	18.86(-32.14)	92.54(+11.25)	14.71(-36.29)	88.27(+6.98)	57.78

Source: Li et al. (2025)

Li, C., Wang, X., Li, M., Miao, B., Sun, P., Zhang, Y., Ji, X., & Zhu, Y. (2025). *Bridging the gap between ideal and real-world evaluation: Benchmarking AI-generated image detection in challenging scenarios*. arXiv. https://doi.org/10.48550/arXiv.2509.09172

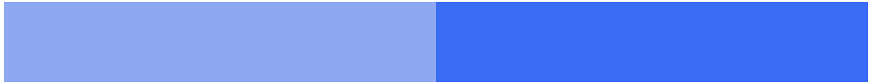
RRDataset: Benchmarking AI Image Detection in the Real World

51,000 images (25.5K real / 25.5K AI-generated) across 7 scenarios, evaluated under 3 real-world conditions

7 Content Scenarios

- **War & Conflict**
- **Disasters & Accidents**
- **Political & Social Events**
- **Medical & Public Health**
- **Culture & Religion**
- **Labor & Production**
- **Everyday Life**

Dataset Composition



25,500 Real Images **25,500 AI-Generated Images**

Li et al. (2025), RRDataset / RRBench — arXiv:2509.09172

3 Real-World Robustness Conditions

Original
Source images

Transmission
Shared via 8 platforms, 2–6 rounds

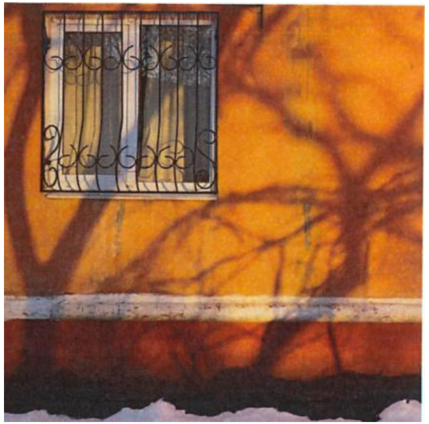
Re-digitization
Scanned / re-photographed

Li, C., Wang, X., Li, M., Miao, B., Sun, P., Zhang, Y., Ji, X., & Zhu, Y. (2025). *Bridging the gap between ideal and real-world evaluation: Benchmarking AI-generated image detection in challenging scenarios*. arXiv. https://doi.org/10.48550/arXiv.2509.09172

RRDataset: Benchmarking AI Image Detection in the Real World

Original | Transmitted | Re-digitization

Real



Fake

Discriminative Low-Level Image Statistics for Real/Fake Separation

Fakes are sharper, more textured, lower-frequency-rich

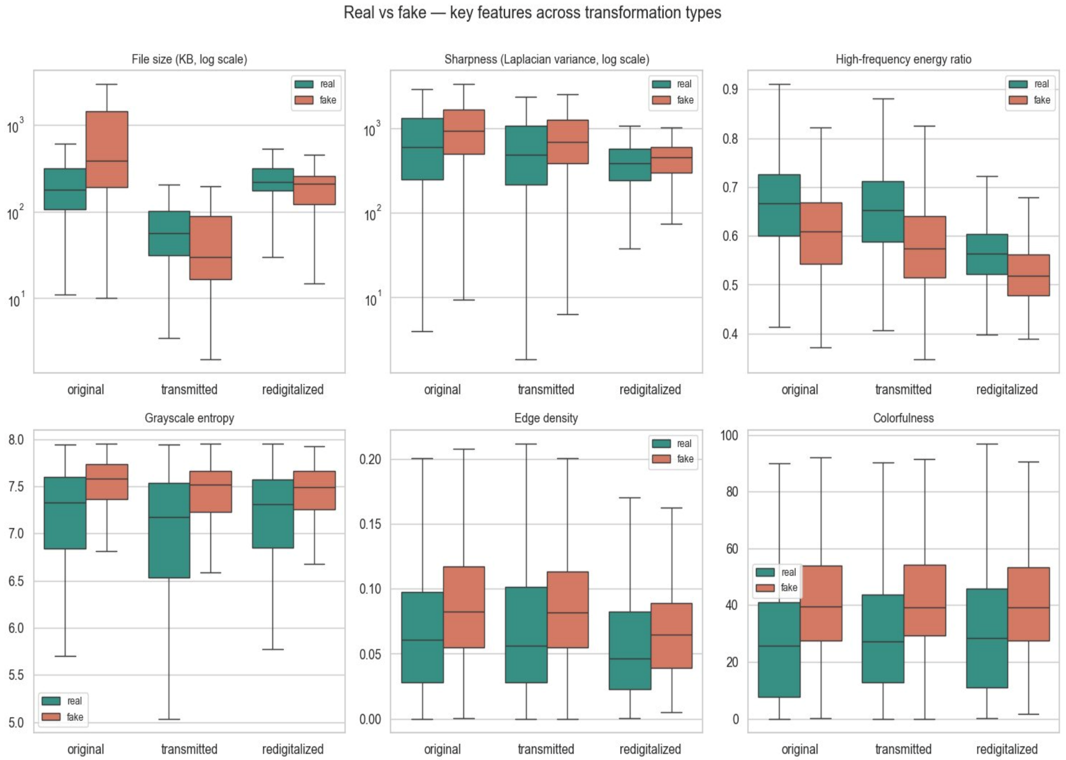
- Fake images show higher sharpness (Laplacian variance), edge density, entropy, and colorfulness than real images
- Real images show higher high-frequency energy ratio - real camera noise vs. generator smoothing

File size is the strongest, most stable separator

- Fakes have ~2x larger files than real images in the 'original' split (log scale)
- The gap reverses for transmitted/redigitalized images - file size alone is not transformation-invariant

Where do we start?

- Since the discriminative features found (file size, sharpness, entropy, high-frequency energy, resolution) are low-level texture/frequency statistics that CNNs naturally capture, and a mid-size CNN seems like a good first step.

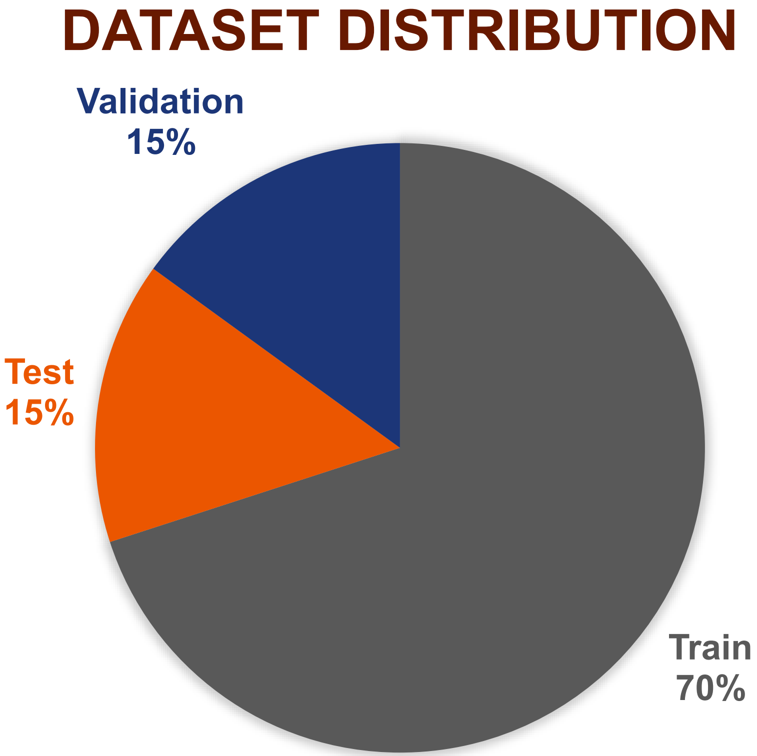


Dataset Description: Images used for model development

For our training we used a subset of 7000 thousand images from the RRDataset to train in one of our local machines

	Original	Transmitted	Redigitalized
Real	1500	1000	1000
Fake	1500	1000	1000

Dataset Description: Dataset Distribution



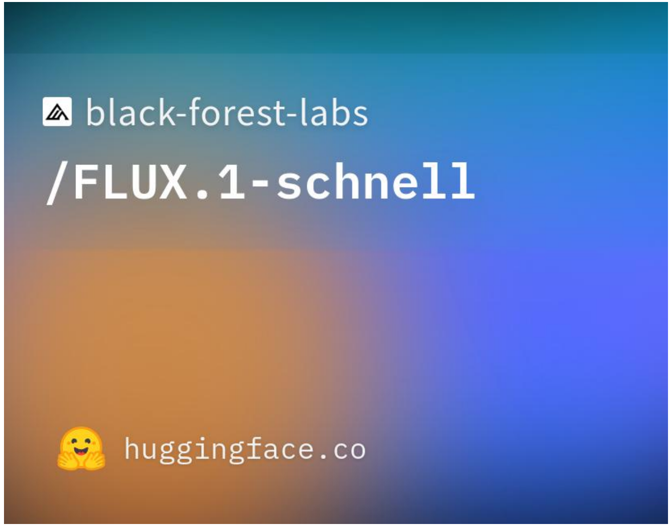
Subset	No. of Images
Train	4813
Validation	1044
Test	1043

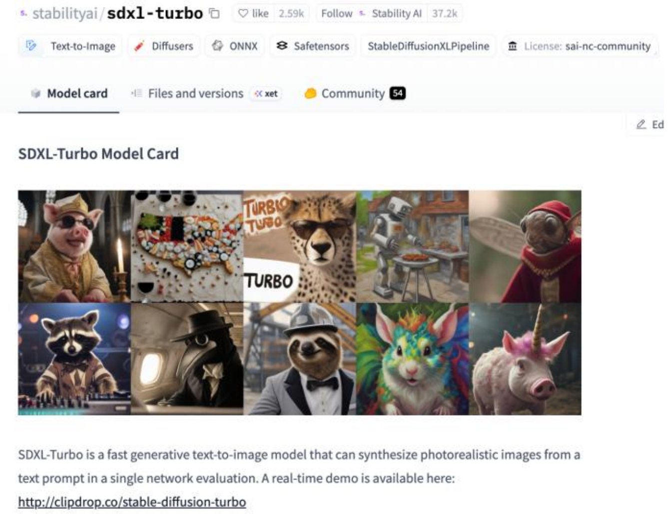
2Fast2Detect: Creating our own dataset

We decided to create our own subset for generalization testing

	Original	Transmitted	Redigitalized
Real	150	150	150
Fake	150	150	150

Creating our Own Dataset: Fake Images





Black Forest Labs. (2024). *FLUX.1-schnell* [Text-to-image model]. Hugging Face. https://huggingface.co/black-forest-labs/FLUX.1-schnell

Sauer, A., Lorenz, D., Blattmann, A., & Rombach, R. (2025). Adversarial diffusion distillation. In A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler, & G. Varol (Eds.), Computer Vision – ECCV 2024 (Lecture Notes in Computer Science, Vol. 15144, pp. 87–103). Springer. https://doi.org/10.1007/978-3-031-73016-0_6

Creating our Own Dataset: Real Images

COCO DATASET (VAL 2017)

Source: COCOdataset.org (2017)

COCO Consortium. (n.d.). COCO: Common Objects in Context. Retrieved June 13, 2026, from https://cocodataset.org/#home

Creating our Own Dataset: Transmision

Original

Source: https://pngimg.com/image/20360

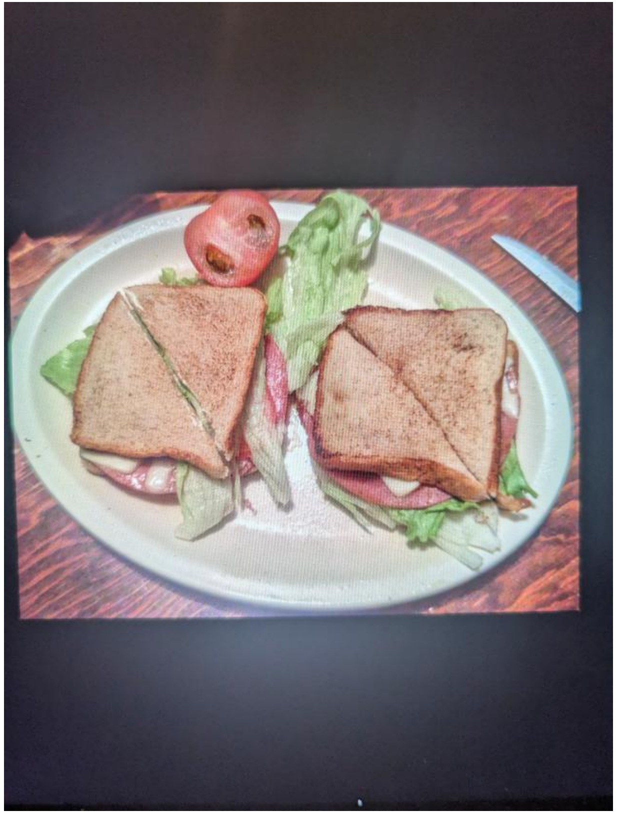
Transmitted

Creating our Own Dataset: Re-digitalization

Methodology:
Take picture of image laptop screen using cellphone camera

Cellphones used:
Apple Iphone 15
Sensor: 48MP Main: 26 mm, ƒ/1.6 aperture

Google Pixel 7
Sensor: 50MP 24 mm, ƒ/1.85 aperture

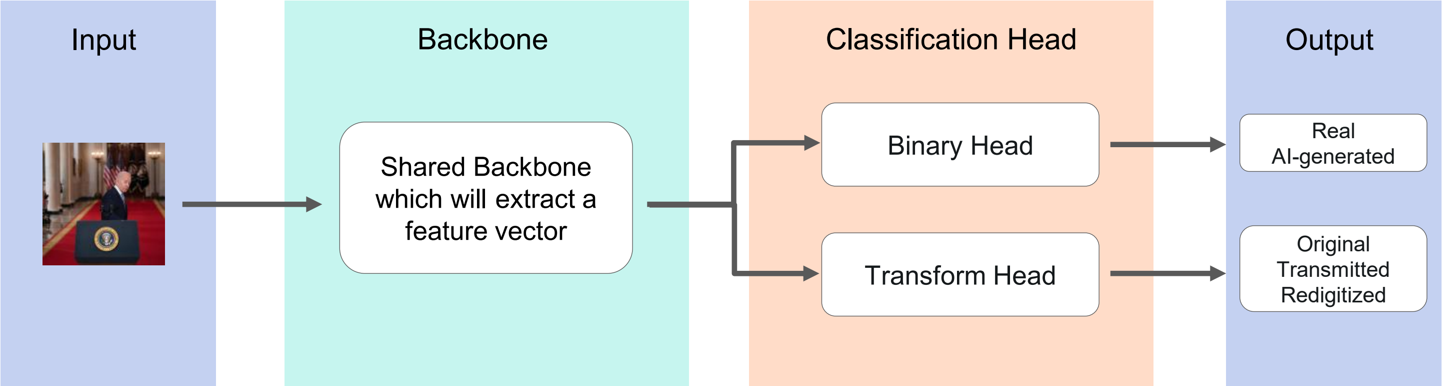


Creating our Own Dataset: Examples

	Original	Transmitted	Re-digitization
Real			
Fake			

Proposed Method

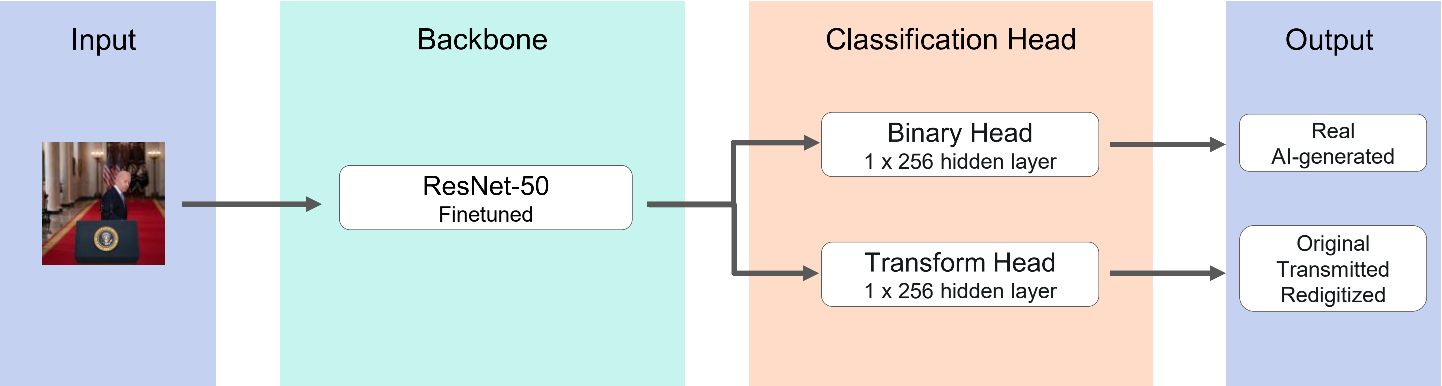
- We want to build a single multi-task model learning both jointly from shared features.
- Two lightweight MLP heads: binary 2-class and transform 3-class
- Joint loss: **L = w1·CE_binary + w2·CE_transform**, class-weighted



Heads are deliberately **small** — the shared trunk does the heavy lifting. Loss weights w1/w2 are ablated.

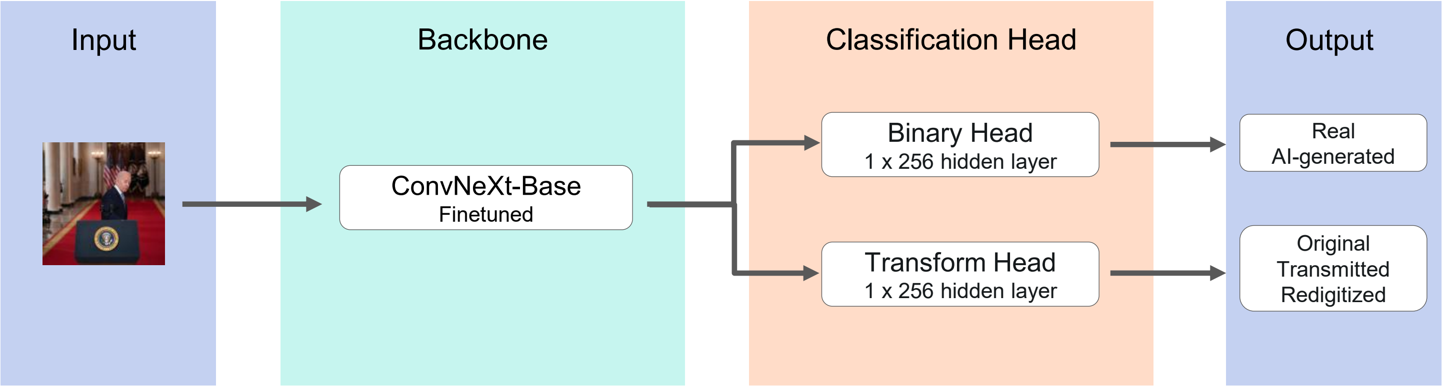
Proposed Method

- Experiments with Shared Backbone



Proposed Method

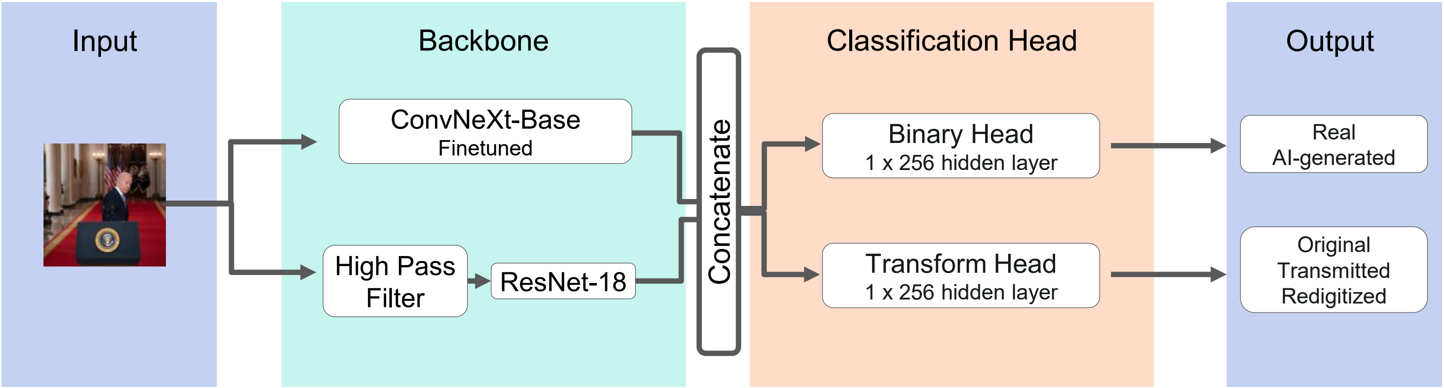
- Experiments with Shared Backbone



Other backbones were also tested

Proposed Method

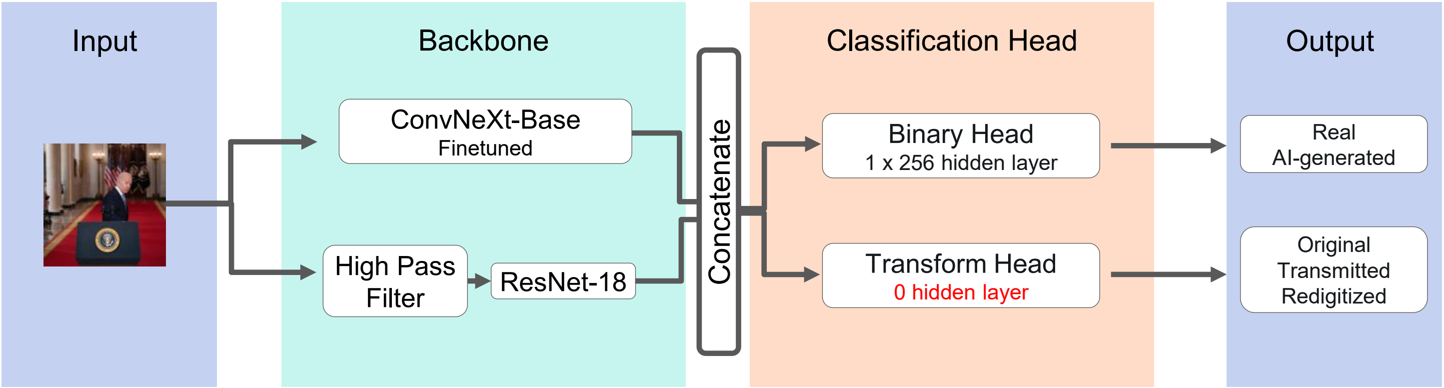
- Extend backbone with Dual Stream



Insight: the real-vs-AI signal lives in **high-frequency generation artifacts**, not image content [Source]

Proposed Method

- Extend backbone with Dual Stream

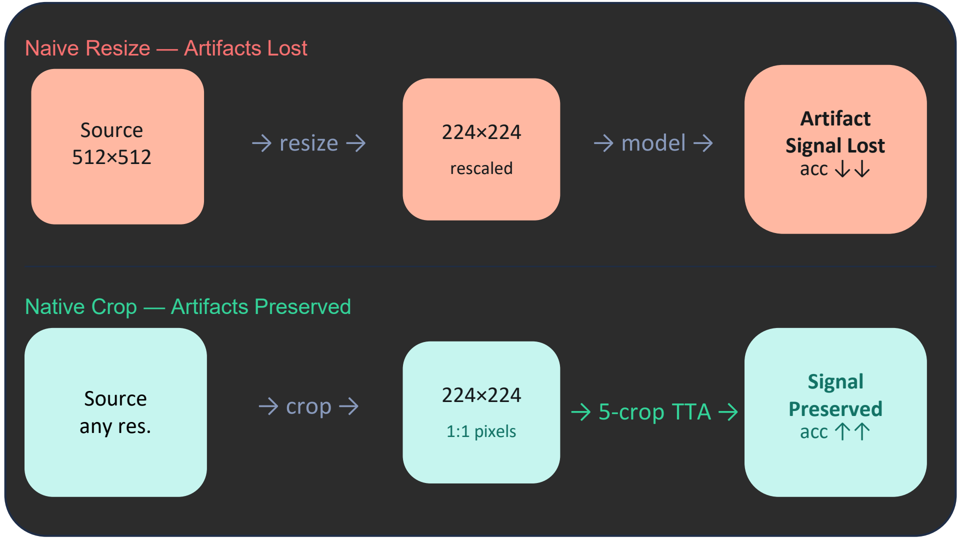


Insight: the real-vs-AI signal lives in **high-frequency generation artifacts**, not image content [Source]

Proposed Method

Forensic Aware Preprocessing

- Standard resize and augmentation **resamples away** the artifacts that will help
- Our choices: **native-pixel crops** (crop, don't downscale) + Flip
- **Test-time augmentation:** average predictions over **5 deterministic crops**
- The transform label *is* a low-level artifact. Downscaling at eval time **destroyed it**. Matching scale at native resolution recovered performance.



Experimental Setup & Evaluation

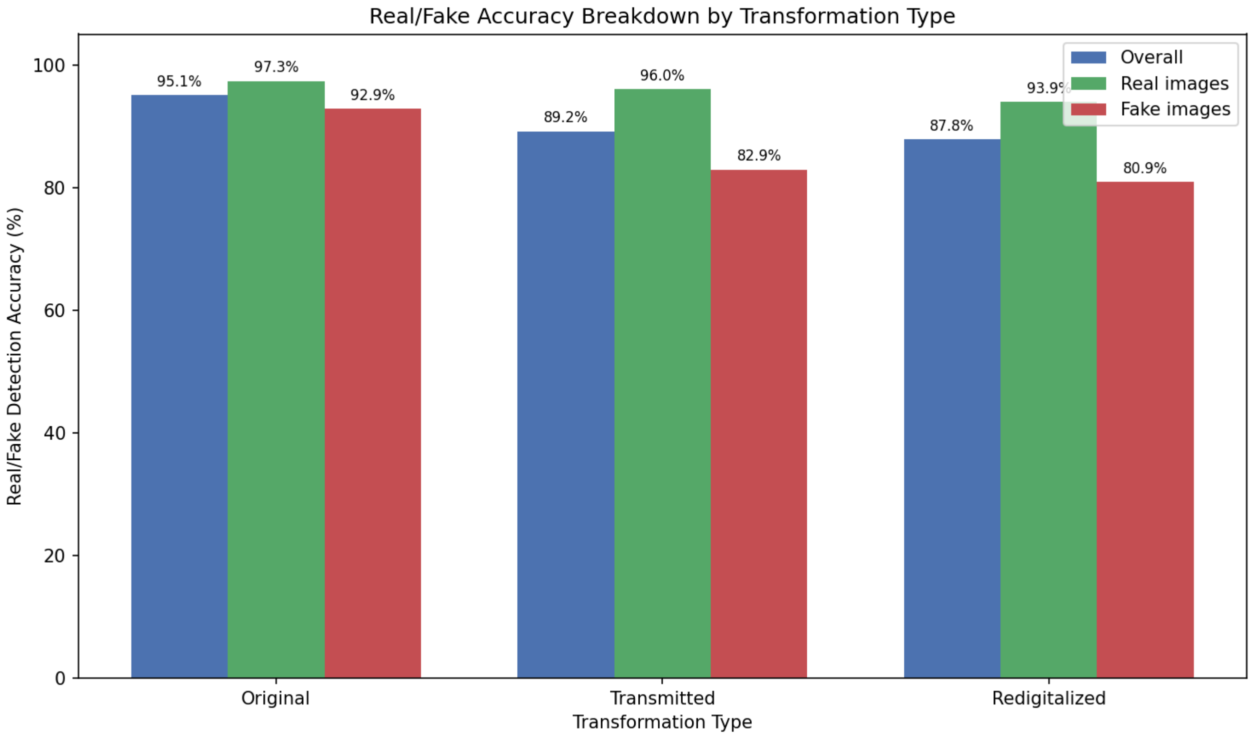
Training Setup

Backbone	ConvNeXt-Base
Input size	224 × 224
Epochs	10
Batch size	32
Optimizer	AdamW
Learning rate	1e-4
LR schedule	Cosine
Early stop	Val F1
Grad clipping	enabled
Loss	Class-weighted CE

Results

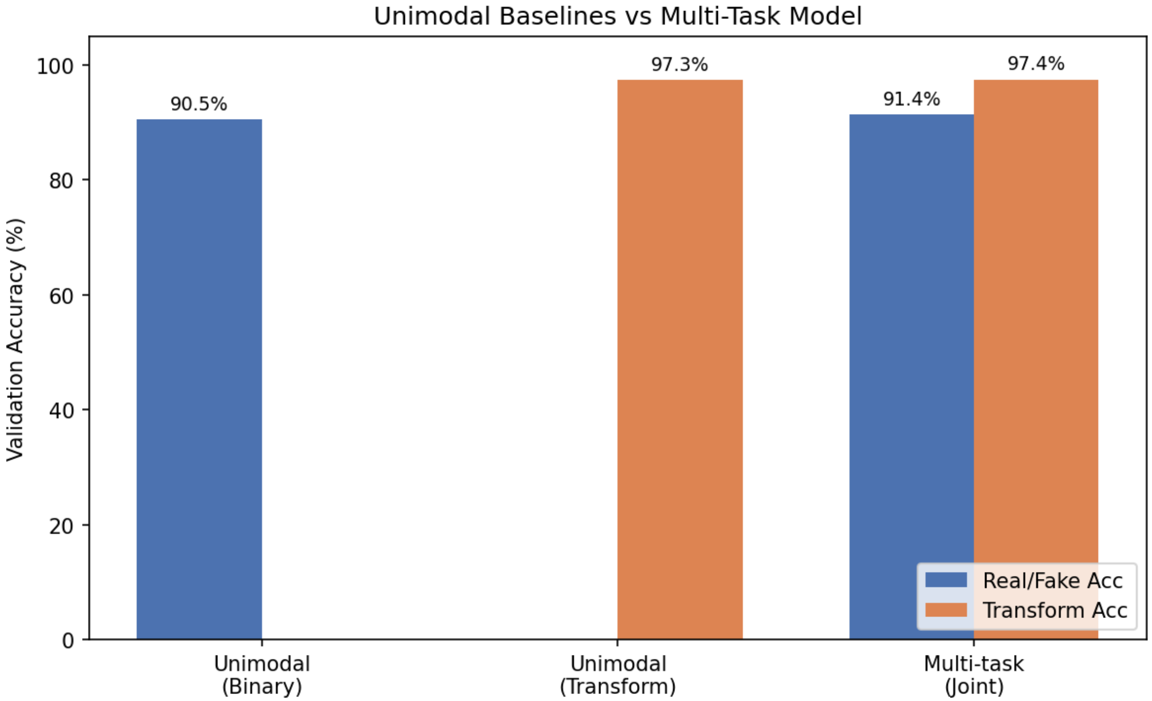
Results
ResNet50

Overall accuracy: 91.4%



Results

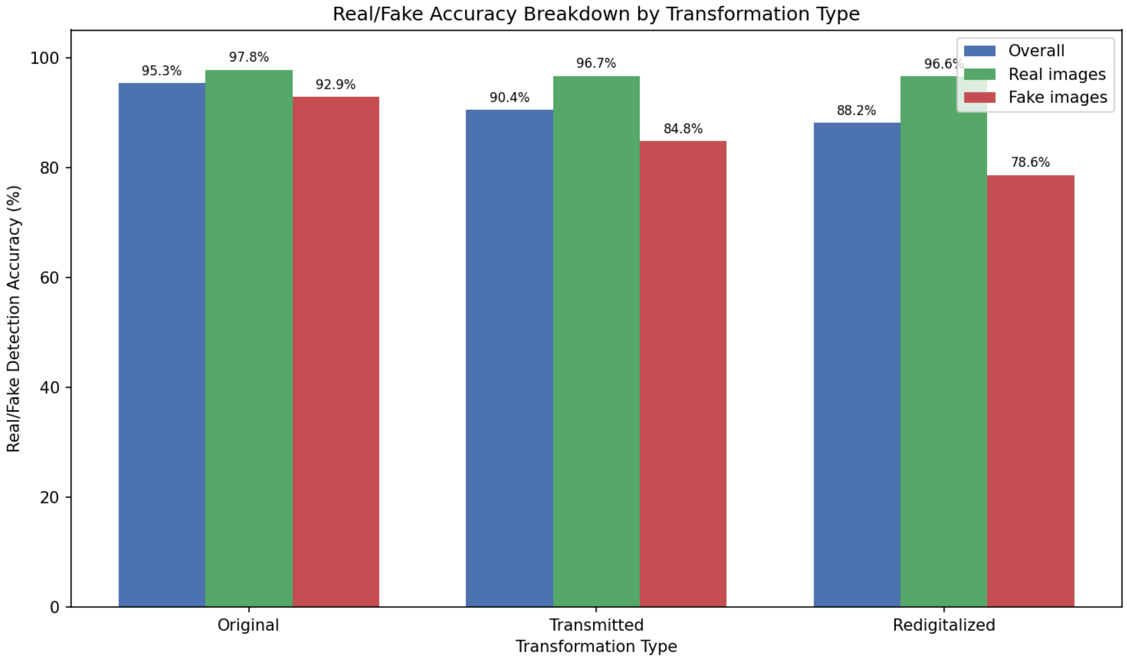
ResNet50



Results

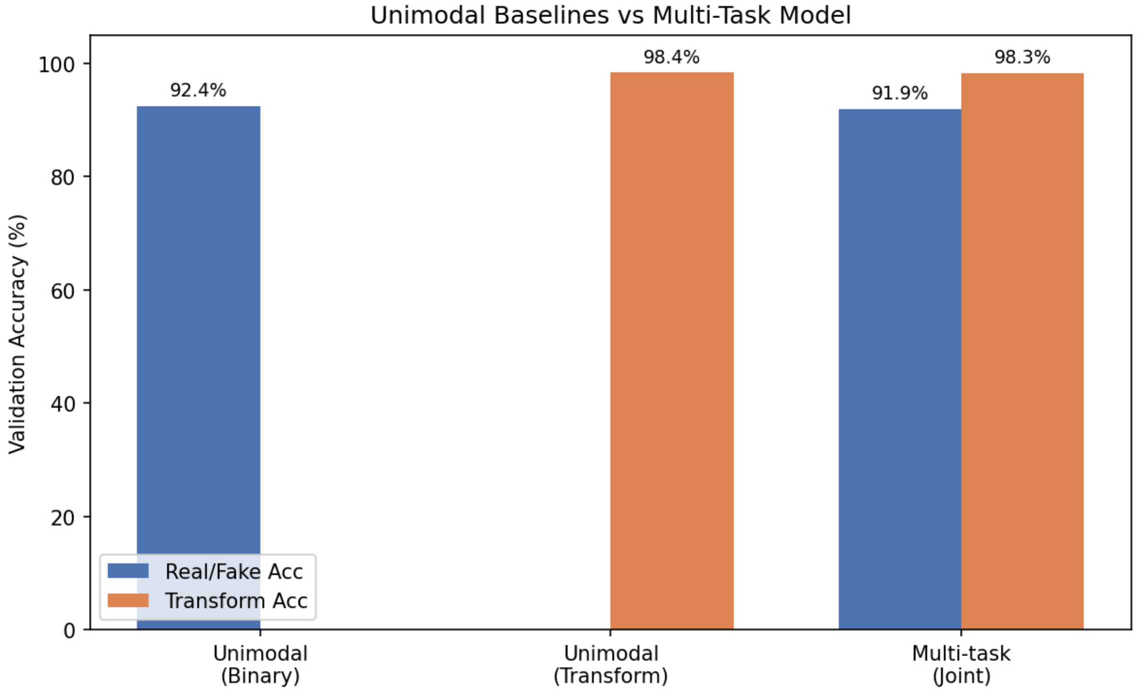
ConvNeXt-Base single-stream

Overall accuracy: 91.9%



Results
ConvNeXt-Base single-stream

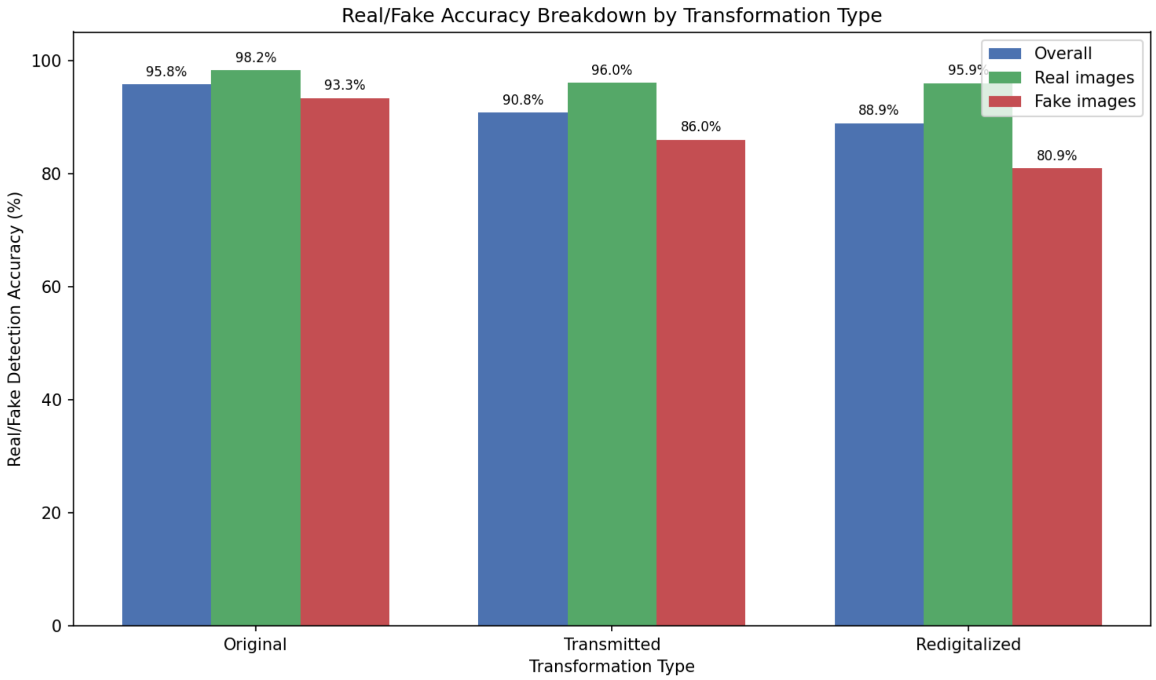
Slightly decreased performance compared to unimodal baseline



Results

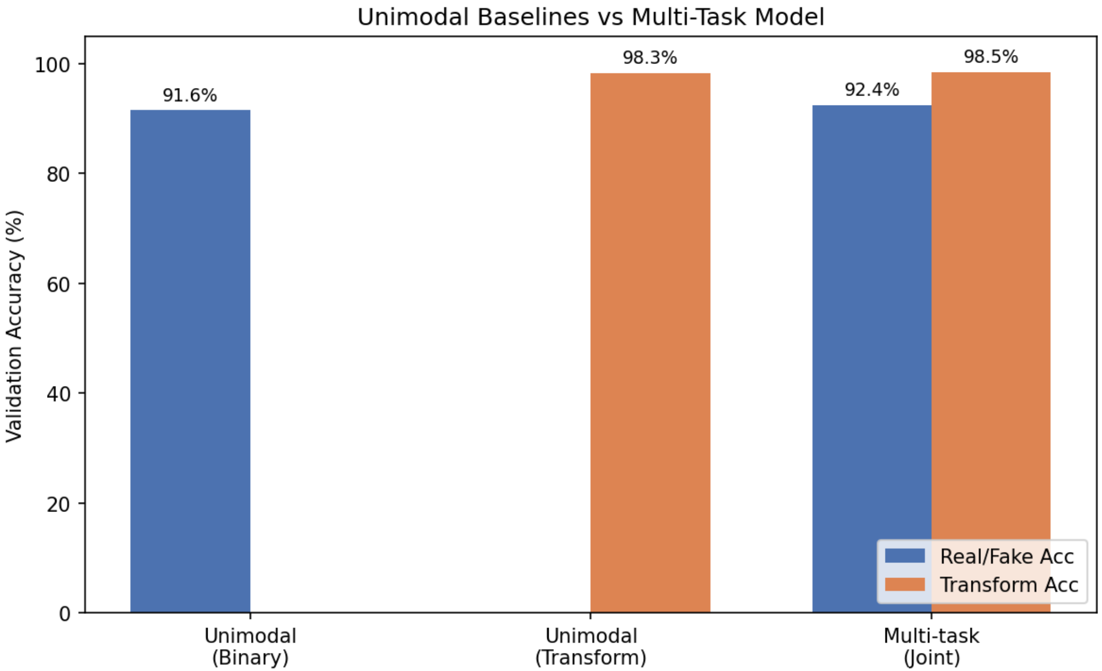
ConvNeXt-Base dual-stream

Overall accuracy: 92.4%

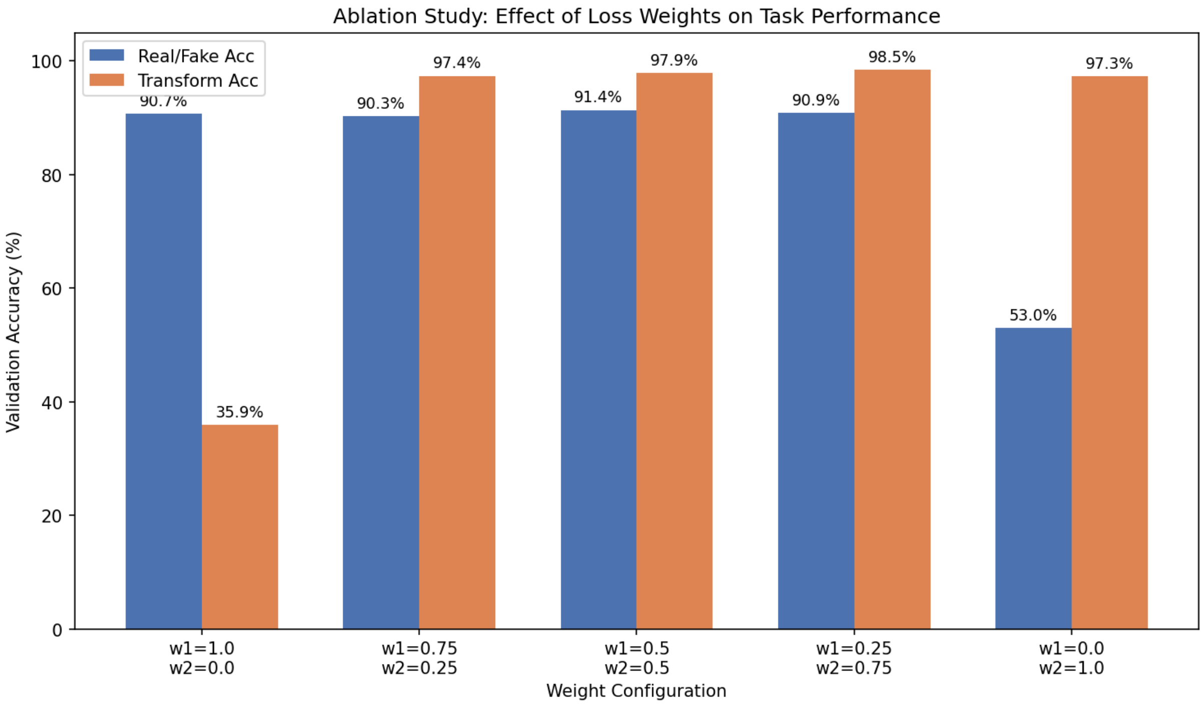


Results

ConvNeXt-Base dual-stream

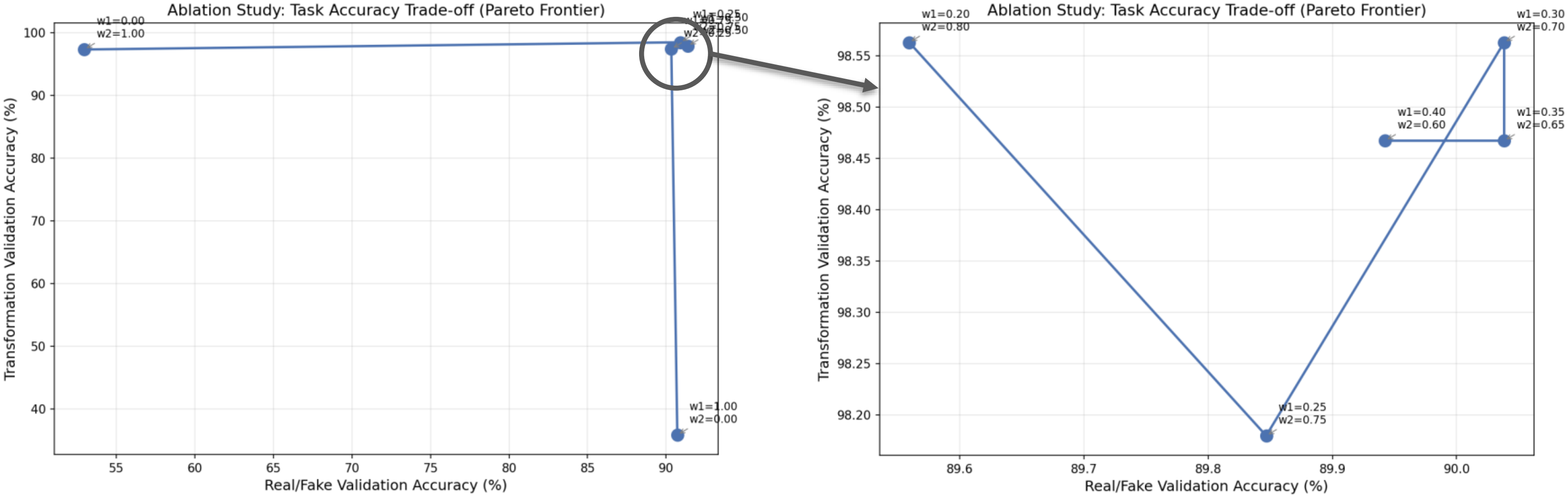


Results
ConvNeXt-Base dual-stream Ablation



Results

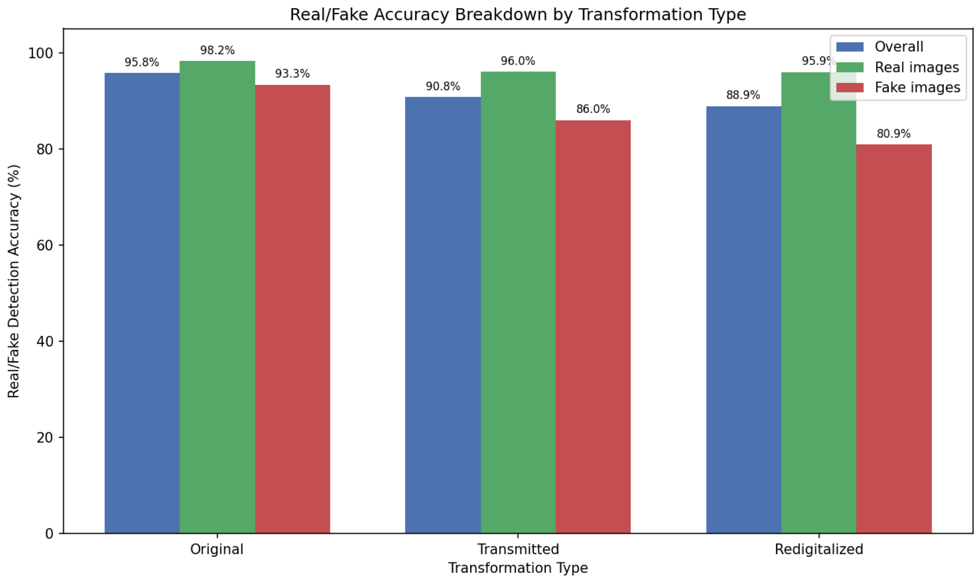
ConvNeXt-Base dual-stream Ablation



Results
Discussion

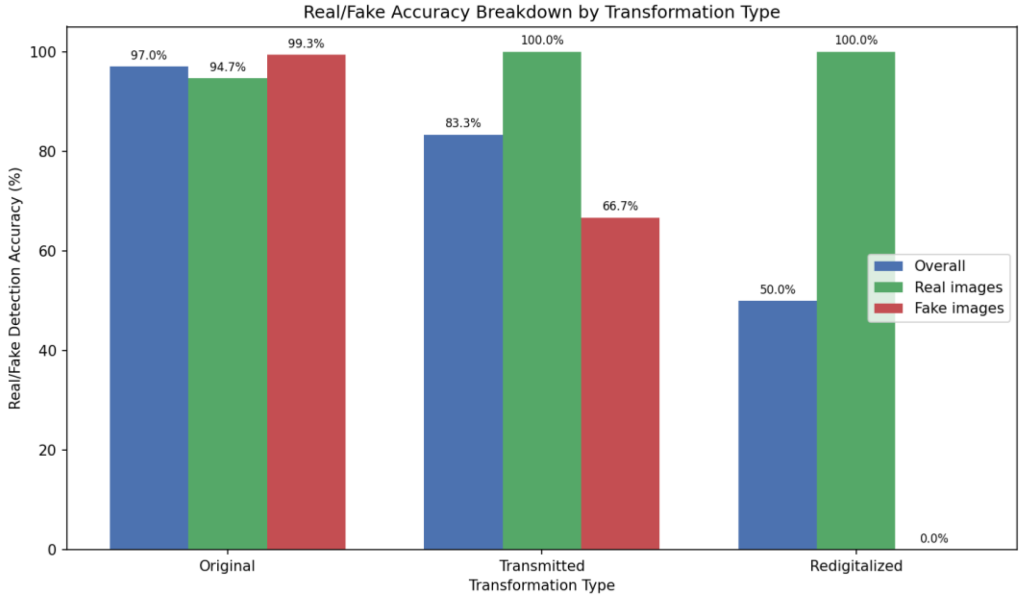
- Multi-task models show considerable potential
- Low re-digitalization performance
- Dual-stream only slightly improves results
- Cropped patches increases performance

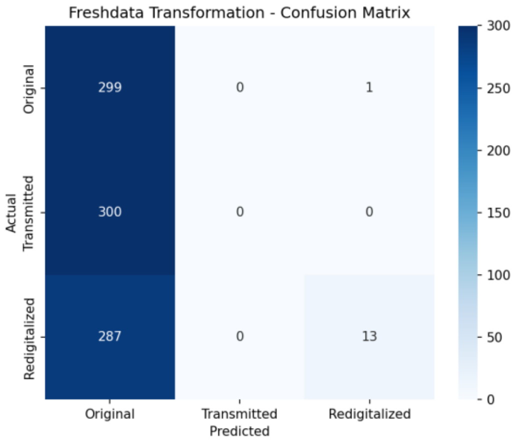
	MT binary	MT transform	Unimodal binary	Unimodal transform
ResNet50	0.9137	0.9741	0.9051	0.9732
ConvNeXt-Base Single-stream	0.9195	0.9828	**0.9243**	0.9837
ConvNeXt-Base Dual-stream	**0.9243**	**0.9847**	0.9156	**0.9856**

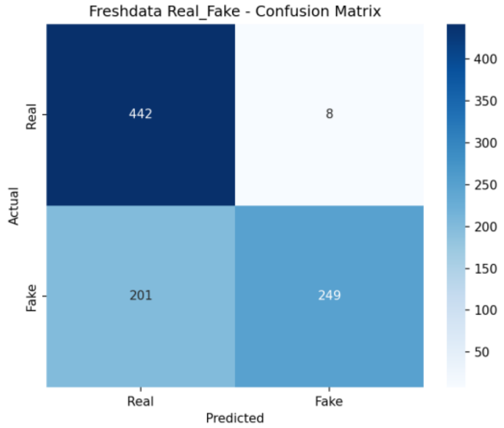


Results
2Fast2Detect

- Doesn't generalize outside of dataset
  - Files are much larger
  - Images out of distribution







Future work

- Improve selection of patches
- Alter amount of patches
- Different patch classification combination approaches
- Expand dataset

Conclusions

- Multi-task training shows considerable potential
- Adding noise pipeline to the branch didn’t give dramatically different results.
- Re-Digitization is the hardest to distinguish between AI and Real
- Lacks real-world generalization capacity

Thank you for your attention!